

VEKTOLUX REAL ESTATE ESCROW ENGINE

Multi-Tier Real Estate Escrow, Anti-Bypass Viewing Passes, Caution Deposit Vault & Land Milestones

Platform: Vektolux Sierra Leone | Standard Currency: SLE (New Leones) | Payment Integrations: Orange Money & Africell Afrimoney

Legal Compliance: OARG (Roxy Building) & MLHCP Cadastral Standards | Version 1.0 (Production Release)

1. Executive Summary & Fintech Topology

Real estate transactions in Sierra Leone (Freetown, Bo, Kenema, Makeni, Waterloo) face chronic systemic challenges:

- **Agency Bypass (Disintermediation):** Clients obtain property locations and bypass listing agents to deal directly with landlords, depriving agents of viewing and finder fees.
- **Caution Deposit Non-Refunds:** Landlords routinely withhold tenant caution deposits post-tenancy without itemized proof of damage.
- **Short-Stay Guesthouse Fraud:** Hosts misrepresent amenities; guests arrive to find uninhabitable conditions with no recourse once funds are sent via Mobile Money.
- **Land Purchase Multiple-Sale Scandals:** Multiple buyers purchase the same parcel due to forged survey plans and unverified title registrations at the Office of the Administrator & Registrar General (OARG).

The **Vektolux Real Estate Escrow System** provides cryptographic anti-bypass protection, algorithmic milestone vesting, and balance-isolated double-entry settlement natively integrated with **Orange Money** and **Africell Afrimoney**.

2. Core Workflows & Financial Formulations

Workflow A: Inspection Pass (Agent-Guided Anti-Bypass System)

Anti-Bypass Guard: Exact property compound addresses, GPS coordinates, and landlord contact details are strictly masked in public listings until an in-person mutual verification occurs.

- **Viewing Fee:** Client books a viewing tour by paying a micro-escrow fee (**SLE 50 - 100**).
- **Cryptographic Token:** Vektolux generates a time-expiring QR code and a 4-digit OTP via HMAC-SHA256.
- **Mutual Verification:** The field agent and client meet at the site; agent scans the client's QR code or client inputs the agent's OTP.
- **Instant 85/15 Disbursal:**

- Platform Technology Cut: **15%** (e.g. SLE 15.00 on SLE 100)
- Agent Net Disbursal: **85%** (e.g. SLE 85.00 on SLE 100)
- **Unmasking Event:** Full landlord contact and exact boundary coordinates are unlocked for the client.

Workflow B: Short-Term Guest Stays & Guesthouses

- **Upfront Inflow:** Gross = (Nightly Rate × Nights) + Refundable Caution Deposit (SLE 500 - 2,000).
- **The 24-Hour Check-in Rule:** Nightly rent is held in escrow for exactly 24 hours after check-in. If no critical defect dispute is raised within 24 hours, net rent (90%) is disbursed to the host and 10% platform fee is realized.
- **Caution Deposit Vault:** Caution deposit remains locked until check-out inspection verifies no missing items or damaged appliances. Clean check-out triggers 100% automated refund to client via Orange Money / Africell / Wallet.

Workflow C: Long-Term Tenancy & Leases (6-12+ Months)

- **Upfront Custody Inflow:** Gross = Agreed Annual Rent + 10% Statutory Agency Commission + 1 to 2 Months' Caution Deposit.
- **Settlement Distribution:**
 - Landlord receives base rent (minus agreed letting cut).
 - Listing agent receives net commission (85% of the 10% agency fee; 15% deducted as Vektolux tech fee).
 - Caution deposit remains locked in the Vektolux Escrow Vault for the full 365-day lease tenure until joint exit inspection.

Workflow D: Land & House Outright Purchases (10 / 40 / 50 Milestones)

Stage	Milestone Name	% Allocation	Release Verification Gate
M1	Title Search Deposit	10%	Certified OARG Search Report (Roxy Bldg, Walpole St) confirming clean root of title.
M2	Survey & Deed Verification	40%	Licensed Surveyor beacon check + Ministry of Lands (MLHCP) Cadastral confirmation.
M3	Final Conveyance & Registry	50%	Signed Deed of Conveyance + OARG Volume/Page registration endorsement.

3. Relational Database Schema Specification

Entity Table	Storage Domain	Key Fields & Invariants
re_inspection_passes	Anti-Bypass Tours	propertyId, clientId, agentId, tourFee, platformFee, agentNetFee, qrHash, otpCode, status
re_escrow_contracts	Master Escrow Vault	contractCode, contractType, grossAmount, cautionDeposit, platformFee, agentCommission, netBeneficiaryExpected, currentState, stay24hReleaseTime
re_escrow_milestones	Land Milestones	contractId, milestoneIndex (1,2,3), percentage, amount, proofDocuments, state, releasedAt
re_caution_deposits	Deposit Custody	contractId, originalDeposit, heldAmount, deductionClaims, refundStatus
re_escrow_disputes	Arbitration	contractId, claimantId, reason, claimedCost, evidenceUrls, status, arbitratorNotes, awards

4. Milestone State Machine

Transition	Trigger Event	Verification Condition	Ledger Action
CREATED → FUNDS_LOCKED	Telco Webhook	Orange/Africell deposit confirmed	Debit Telco Liability; Credit Escrow Vault
FUNDS_LOCKED → AGENT_DISPATCHED	Agent Tour Acceptance	Agent assigned & geo-fenced	Push notification sent to client
AGENT_DISPATCHED → MILESTONE_VERIFIED	Mutual QR / OTP Scan	Client & agent GPS match	Disburse 85% to Agent, 15% to Platform
FUNDS_LOCKED → CHECKED_IN	Guest Door QR Scan	Short-stay check-in verified	Arm 24h dispute hold timer
CHECKED_IN → FULLY_SETTLED	24h Timer Cleared	No defect dispute logged	Disburse net rent to host
ANY → UNDER_ARBITRATION	Dispute Filed	Damage/defect evidence submitted	Freeze payouts & caution deposit

5. REST API Specification (OpenAPI 3.0 Standard)

- POST /api/v1/real-estate/inspection-pass/initiate — Books viewing tour, locks micro-escrow, returns QR & OTP.
- POST /api/v1/real-estate/inspection-pass/verify — Validates mutual QR/OTP, unmask contact, disburses 85% to agent.
- POST /api/v1/real-estate/escrow/initiate — Initiates stay, lease, or land purchase contract.
- POST /api/v1/real-estate/escrow/short-stay/check-in — Records guest arrival and arms 24h timer.
- POST /api/v1/real-estate/escrow/caution-deposit/refund — Check-out signoff and caution refund.
- POST /api/v1/real-estate/escrow/land-milestone/verify — Releases 10% / 40% / 50% land milestones.

6. Dispute Resolution Framework

Tier 1: Direct In-App Conciliation (48-Hour SLA)

Host and tenant have 48 hours to agree on damaged items or cleanliness issues.

Tier 2: Vektolux Independent Loss Adjuster & Admin Adjudication

If unresolved after 48 hours, Vektolux adjudicators review pre-inspection and post-inspection photos, receipts, and contractor quotes in Freetown. Legitimate damage costs are deducted from caution deposit; remainder is refunded to tenant immediately.